Tristan Ryoma Fuchs

フックストリスタン龍馬

Master's student (M2)
Department of Physics, The University of Tokyo
Arita Group · MERIT-WINGS program student

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RESEARCH INTERESTS

Superconductivity · first-principles (DFT) calculations

EDUCATION

Apr 2025 – present **Master's Program in Physics, The University of Tokyo**
Supervisor: Ryotaro Arita.

Apr 2021 – Mar 2025 **B.Sc. in Physics, Department of Physics, Faculty of Science, Tohoku University**
Supervisor: Takashi Koretsune.
First-principles Migdal–Eliashberg calculation of the superconducting gap functions of Al and H_3S.

Sep 2016 – Jun 2020 **Shanghai Jincai High School, International Division, Shanghai, China**
Grades 9–12; International Baccalaureate Diploma Programme (Grades 11–12).

Apr 2008 – Aug 2016 **Shanghai Japanese School, Pudong Campus, Shanghai, China**
Grades 1–9.

PUBLICATIONS

1. **T. R. Fuchs**, T. Nomoto, H. Watanabe, and R. Arita, "Microscopic calculation of coherence lengths and magnetic penetration depth in multiband superconductors," arXiv:2609.06014 [cond-mat.supr-con] (2026), under review.

PRESENTATIONS

International Conferences

Dec 2026 **Microscopic Calculation of Characteristic Length Scales in Multi-Band Superconductors** POSTER SCHEDULED
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — International Quantum Materials Symposium (i-QMS 2026), Jeju, Korea

Nov 2026 **Development of Microscopic Methods for Calculating Characteristic Length Scales in Multiband Superconductors** POSTER SCHEDULED
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — CDS × QUAST Joint Workshop: Frontiers in Correlated Quantum Matter, The University of Tokyo, Tokyo, Japan

Nov 2026 **Coherence Lengths and Magnetic Penetration Depth in Multiband Superconductors from Microscopic Calculations** POSTER SCHEDULED
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — International Workshop on Frontiers in Ab-initio Many-Body Interactions (FAMBI 2026), Sendai, Japan

Nov 2026 **Coherence Lengths and Magnetic Penetration Depth in Multiband Superconductors Based on Microscopic Calculations** POSTER SCHEDULED
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — 27th Asian Workshop on First-Principles Electronic Structure Calculations (ASIAN-27), Kaohsiung, Taiwan

Sep 2026 **Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Multi-Band Superconductors** POSTER
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — International Conference on Strongly Correlated Electron Systems (SCES 2026), Toyama, Japan

Sep 2026 **Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Multi-Band Superconductors** POSTER
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — International Conference on Asymmetric Quantum Matters (Asymmetry 2026), Toyonaka, Osaka, Japan

Feb 2026 **Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Two Band Superconductors** POSTER
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — CEMS Topical Meeting on Recent Trends in Superconductivity, RIKEN, Wako, Japan

Sep 2025 **Microscopic Calculation of Coherence Length and Magnetic Penetration Depth in Superconductors** POSTER
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — Autumn School on Correlated Electrons: Understanding Correlated Materials with DMFT, Forschungszentrum Jülich, Germany

Domestic Conferences (Japan)

Sep 2026 — **Calculation of Coherence Lengths and Magnetic Penetration Depth Based on the Real-Space Bogoliubov–de Gennes Method** POSTER
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — 81st Annual Meeting of the Physical Society of Japan (JPS), The University of Tokyo, Komaba Campus (in Japanese)

Sep 2026 — **Microscopic Calculation of Characteristic Length Scales in Multi-band Superconductors Based on the Extended Ginzburg–Landau Method** POSTER BEST PRESENTATION AWARD
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — MERIT Camp 2026 (MERIT-WINGS, The University of Tokyo), Minano, Saitama (in Japanese)

Aug 2026 — **Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Multi-Band Superconductors Based on Extended Ginzburg–Landau Theory** ORAL POSTER
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — 71st Condensed Matter Physics Summer School (物性若手夏の学校), Aso, Kumamoto (in Japanese)

May 2026 — **Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Multi-Band Superconductors** POSTER
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — Kickoff Meeting of the Transformative Research Area (A) "Asymmetric Quantum Matters," Nasu, Tochigi (in Japanese)

Mar 2026 — **Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Two-Band Superconductors** ORAL
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — JPS 2026 Spring Meeting, online (in Japanese)

Mar 2026 — **Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Two-Band Superconductors** POSTER BEST POSTER AWARD
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — Computational Physics Spring School 2026 (計算物理春の学校), Naha, Okinawa (in Japanese)

Jan 2026 — **Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth** ORAL
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — 15th MERIT Colloquium (MERIT-WINGS, The University of Tokyo) (in Japanese)

AWARDS AND FELLOWSHIPS

Sep 2026 — **Best Presentation Award, MERIT Camp 2026**

Mar 2026 — **Best Poster Award, Computational Physics Spring School 2026** (計算物理春の学校)

Oct 2025 – present — **MERIT-WINGS Program Student**
World-leading Innovative Graduate Study Program for Materials Research, Information, and Technology (MERIT-WINGS), The University of Tokyo

EXPERIENCE

Dec 2023 – present — **Student Staff Lead, Nagase Brothers Inc. (education company)**
Oct 2024 – present: development of internal automation tools at Yotsuya Otsuka
Dec 2023 – Mar 2026: project management at Toshin Digital University

Spring 2024 — **Teaching Assistant, 学問論 (first-year general education course), Tohoku University**

LEADERSHIP AND ACTIVITIES

Oct 2025 – Sep 2026 — **Deputy Organizer, MERIT Camp** — The University of Tokyo

Oct 2023 – Mar 2025 — **Student Staff, Kita-Aobayama Library (Science & Pharmacy Library)** — Tohoku University

Apr 2022 – Mar 2023 — **Vice President, 物理の会 ("Butsuri no Kai," physics seminar society)** — Tohoku University

CERTIFICATIONS AND TEST SCORES

2026 — **EIKEN Grade 1**
実用英語技能検定（英検）1級

2025 — **Applied Information Technology Engineer Examination**
応用情報技術者試験 — passed

2024 — **TOEFL iBT 105**

2024 — **TOEIC Listening & Reading 960**

2022 — **HSK Level 4**
汉语水平考试 (Chinese Proficiency Test)

LANGUAGES AND SKILLS

Languages: Japanese (native), English (fluent), German (conversational), Chinese (basic)

Programming: Julia (main language for research), Python, Google Apps Script

Last updated: October 2026